

Topology PhD Exam, September 1993

Answer the following questions without giving proofs.

1. Let $X \subset Y$.
 - a) What does it mean for X to be a retract of Y ?
 - b) What does it mean for X to be a deformation retract of Y ?
2. Let K be a finite simplicial complex. Define $\chi(K)$, the Euler characteristic of K .
3. Define what it means for a topological space to be
 - a) normal
 - b) connected
4. State the Brouwer fixed point theorem.

Answer the following questions, giving proofs or counterexamples.

5. Let $p : X \rightarrow Y$ be a d -sheeted covering projection, where X and Y are finite simplicial complexes. Show that $\chi(X) = d \cdot \chi(Y)$.
6. Compute the fundamental group of a closed, orientable surface X of genus 3.
7. Let X be a closed, orientable surface of genus 3.
 - a) Compute the homology groups of X .
 - b) Compute the Euler characteristic of this surface X .
8. Show that every continuous map $f : S^2 \rightarrow S^1$ is homotopic to a constant map.
9. For $i = 1, 2, \dots$, let P_i be a 2-dimensional plane in $\mathbb{R}^3$.
 - a) Show that $S = \mathbb{R}^3 \setminus \bigcup_{i=1}^{\infty} P_i$ is not empty.
 - b) Must S be an open set?
10. Let C be the Cantor set and let S be any compact metric space. Show that there is a continuous map $f : C \rightarrow S$ such that f is onto.